Assignment 1

SAKSHI PARATE - 7073022

Week 1

3. Add Meta-data

How many cells are in each sample?

```
> ncol(seurat1)
[1] 6956
> ncol(seurat2)
[1] 15102
> ncol(seurat3)
[1] 18182
> ncol(seurat4)
[1] 11943
```

How many genes are in the expression matrices?

```
> nrow(seurat1)
[1] 27260
> nrow(seurat2)
[1] 29509
> nrow(seurat3)
[1] 29472
> nrow(seurat4)
[1] 29205
```

What information is now part of the metadata of the objects?

```
> colnames(seurat1@meta.data)
[1] "orig.ident"   "nCount_RNA"   "nFeature_RNA" "Donor"        "Replicate"
[6] "Sex"
> head(seurat1@meta.data)
            orig.ident nCount_RNA nFeature_RNA Donor Replicate Sex
AGCAACCTCGCG   PBMC_556        507         425    C2        T1   F
GTTGATTCCTGA   PBMC_556        276         200    C2        T1   F
TAGTCGTTTGGT   PBMC_556        293         201    C2        T1   F
GGCCCATTATTC   PBMC_556        651         474    C2        T1   F
TAGCCGCACCGG   PBMC_556        347         203    C2        T1   F
TATAGTGTGGAG   PBMC_556        631         447    C2        T1   F
```

Week 2

4.1 Preprocessing

Filtering and Doublet removal are done before merging because filtering removes low quality cells and doublet detection depends on each sample's structure. if we merge first, doublets become harder to detect.

Normalization and Feature selection are done after merging because after merging, global normalization ensures samples are comparable and feature selection must be done on the merged dataset to choose variable genes across all donors.

Filtering of each sample was done using three QC metrics:

1) nFeature_RNA (number of detected genes)
2) nCount_RNA (total RNA counts)
3) percent.mt (percentage of mitochondrial RNA)

Cells with <200 genes were removed because they likely represent empty droplets or low-quality cells.

Cells with >6000 genes were filtered out as potential doublets or multiplets.

Cells with >10% mitochondrial reads were removed because they are typically stressed or dying.

These thresholds follow standard Seurat and 10X Genomics recommendations and ensure only high-quality single cells are retained for downstream analysis.

Doublet removal is needed because in single cell droplet technologies, two cells can accidentally be captured in the same droplet. The RNA from both cells is sequenced together and appears as one fake cell with twice the RNA content and mixed gene expression. These artificial hybrid profiles create incorrect clusters, bias normalization, and lead to wrong cell type predictions. Doublet Finder identifies and removes these artificial cells so downstream analysis reflects only true biological cells.

The optimal value for pK is:

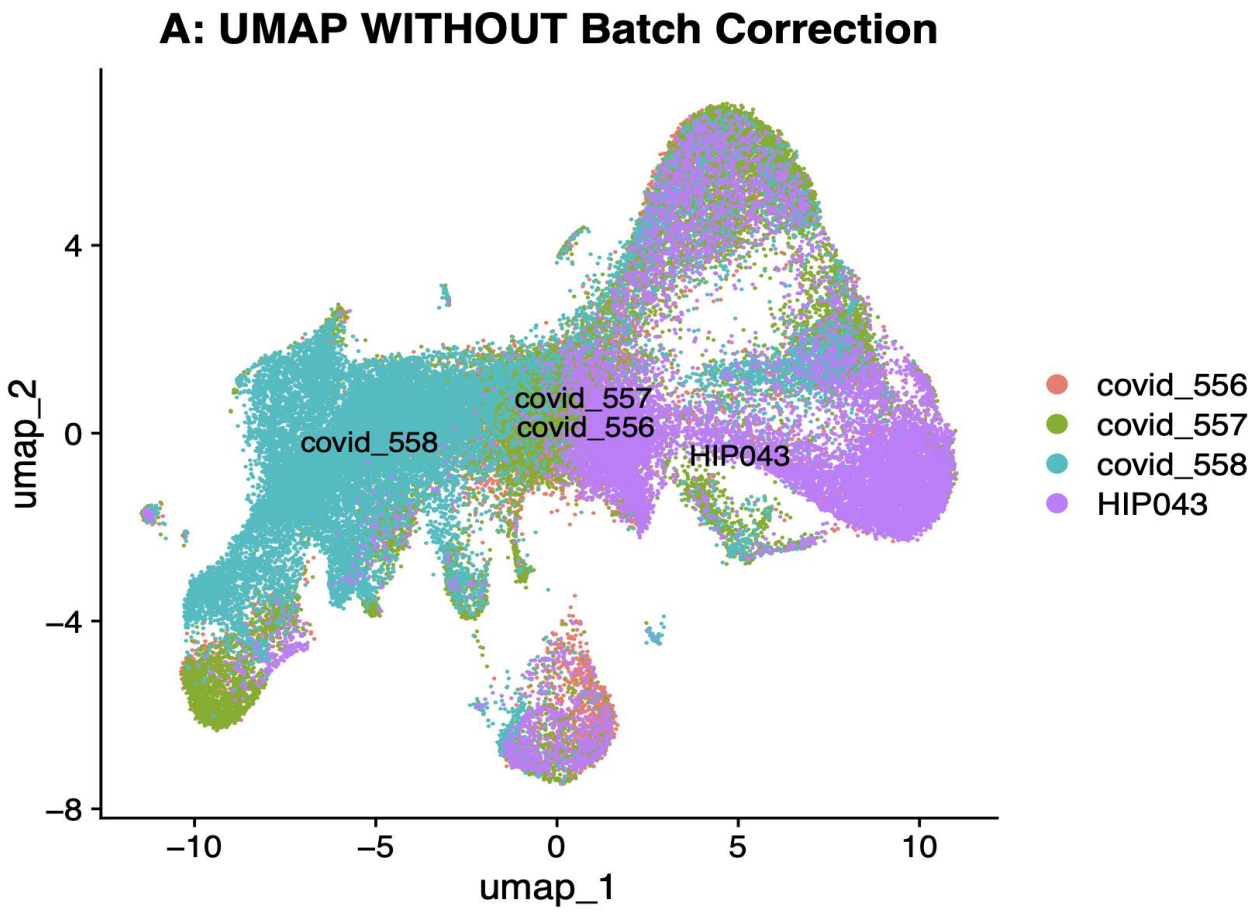
```
$seurat1$pK
[1] 0.08
$seurat2$pK
[1] 0.09
$seurat3$pK
[1] 0.3
$seurat4$pK
[1] 0.3
```

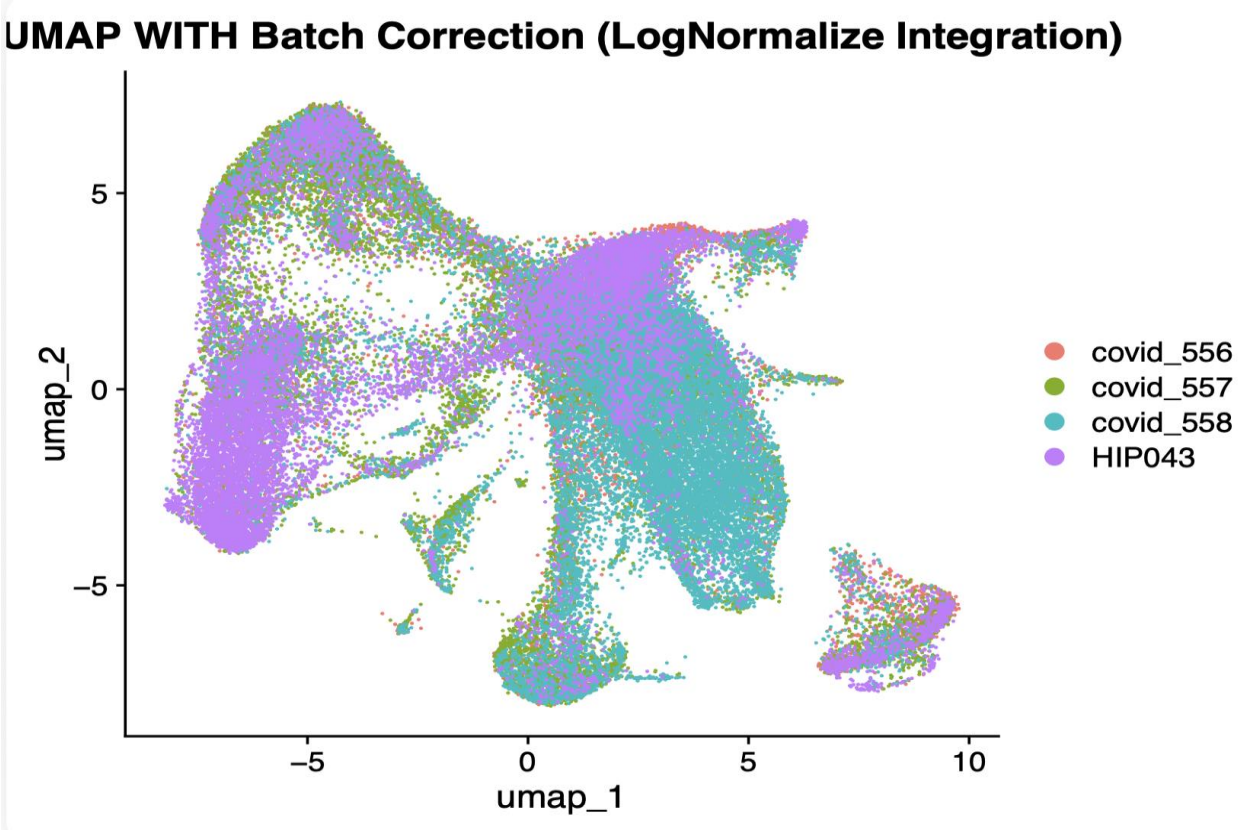
Which Normalization method is used by NormalizeData() by default?
NormalizeData() uses "LogNormalize" by default. That means for each cell it divides each gene count by the total counts in that cell, multiplies by a scale factor (default scale.factor = 1e4), then applies log1p (log(1 + x)). This produces comparable expression values across cells.

What is the purpose of Feature Selection (HVG selection)?
We select highly variable genes (HVGs) to focus downstream analyses (PCA, clustering, integration) on the genes that show meaningful biological variation rather than technical noise. This reduces dimensionality, speeds up computation, and often improves signal for clustering and integration.

How are they selected?
FindVariableFeatures() (default method = "vst") computes per-gene statistics (variance stabilized), ranks genes by standardized variance, and returns the top n genes (e.g. nfeatures = 2000). These selected genes are the HVGs.

4.2 Batch-Correction

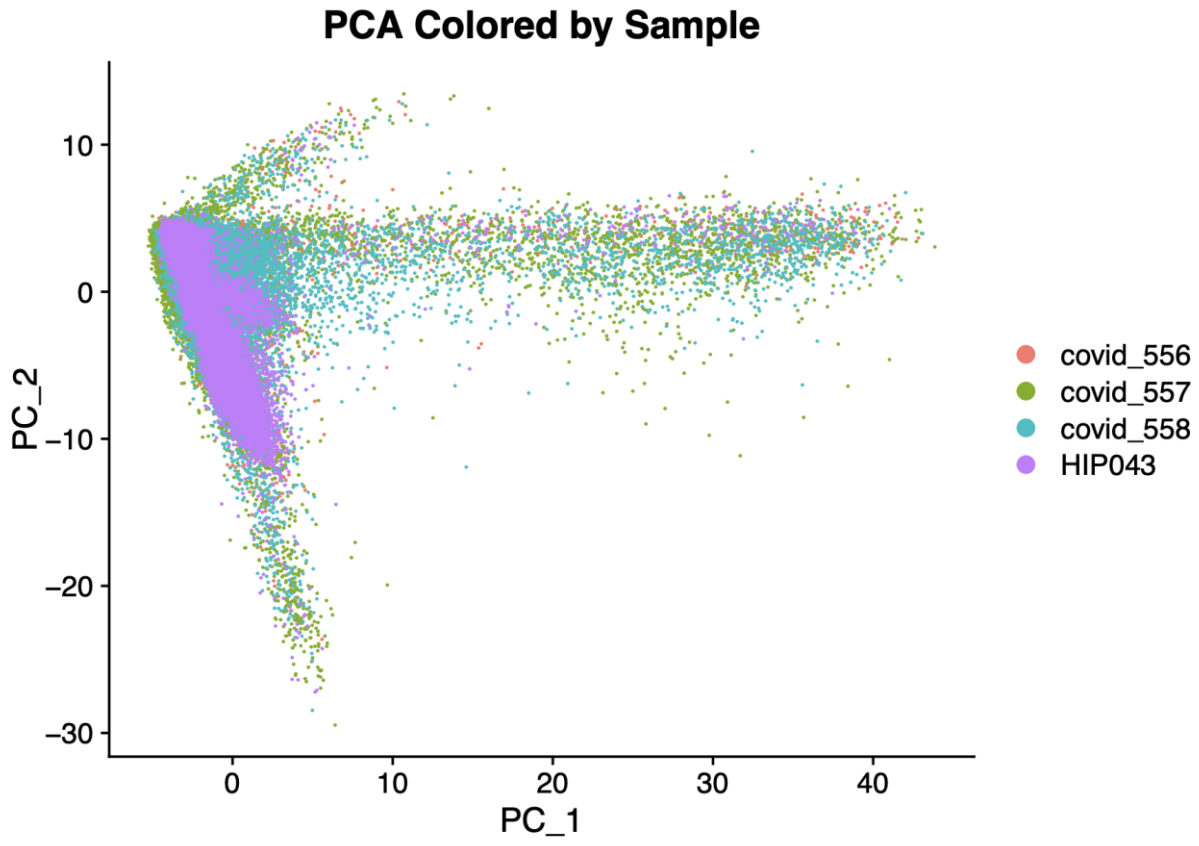


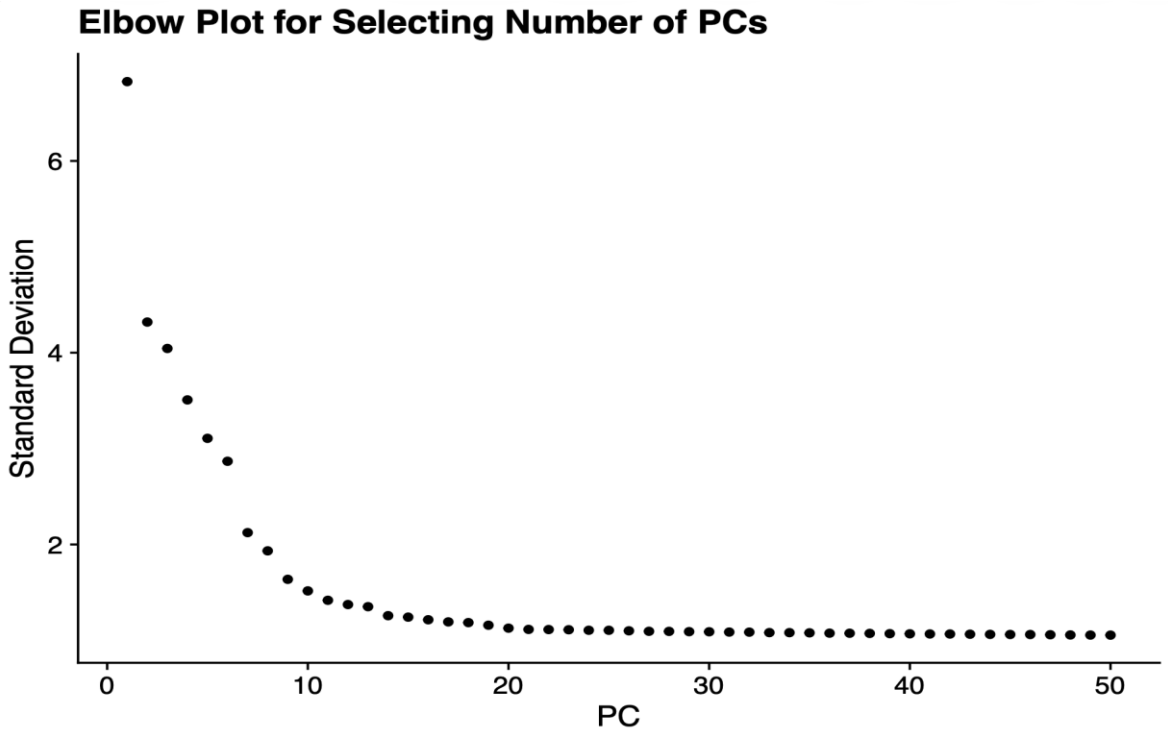


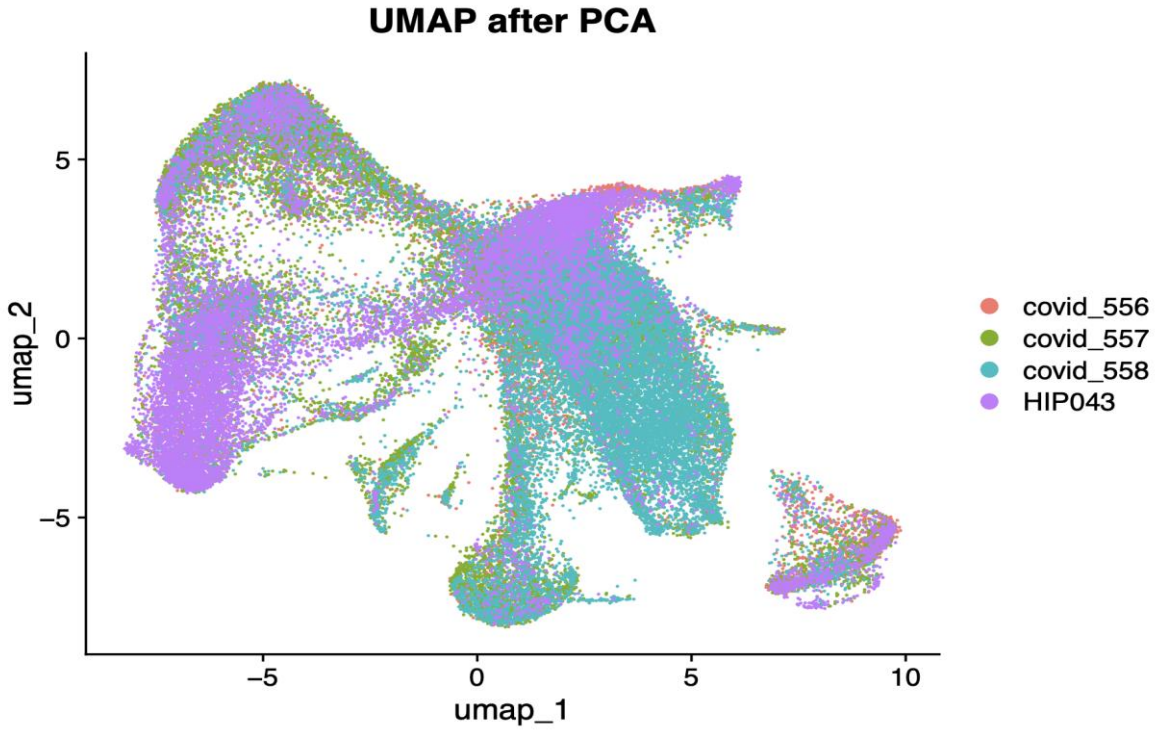
Compare the outcomes of A and B.
Without batch correction (Plot A), the UMAP shows strong sample-driven clustering: cells from covid_556, covid_557, covid_558, and HIP043 occupy separate regions, indicating that technical variation between samples is dominating the embedding.
After batch correction using Seurat's integration workflow (Plot B), the UMAP shows well-mixed samples. The sample labels no longer define the global structure, meaning biological variation (cell types) now drives clustering.

Is Batch-Correction necessary? If yes, name the parameters and explain (with the necessary plots) why a correction for this parameter may be necessary?
Batch correction is therefore necessary. The main parameter requiring correction is the sample identity (orig.ident), which reflects differences in sequencing depth (nCount_RNA), detected genes (nFeature_RNA), and mitochondrial percentage (percent.mt). These parameters vary strongly between samples and produce sample-specific clusters, as visible in Plot A. Integration removes this artificial separation and reveals true biological structure (Plot B).

5.1 Dimensionality Reduction





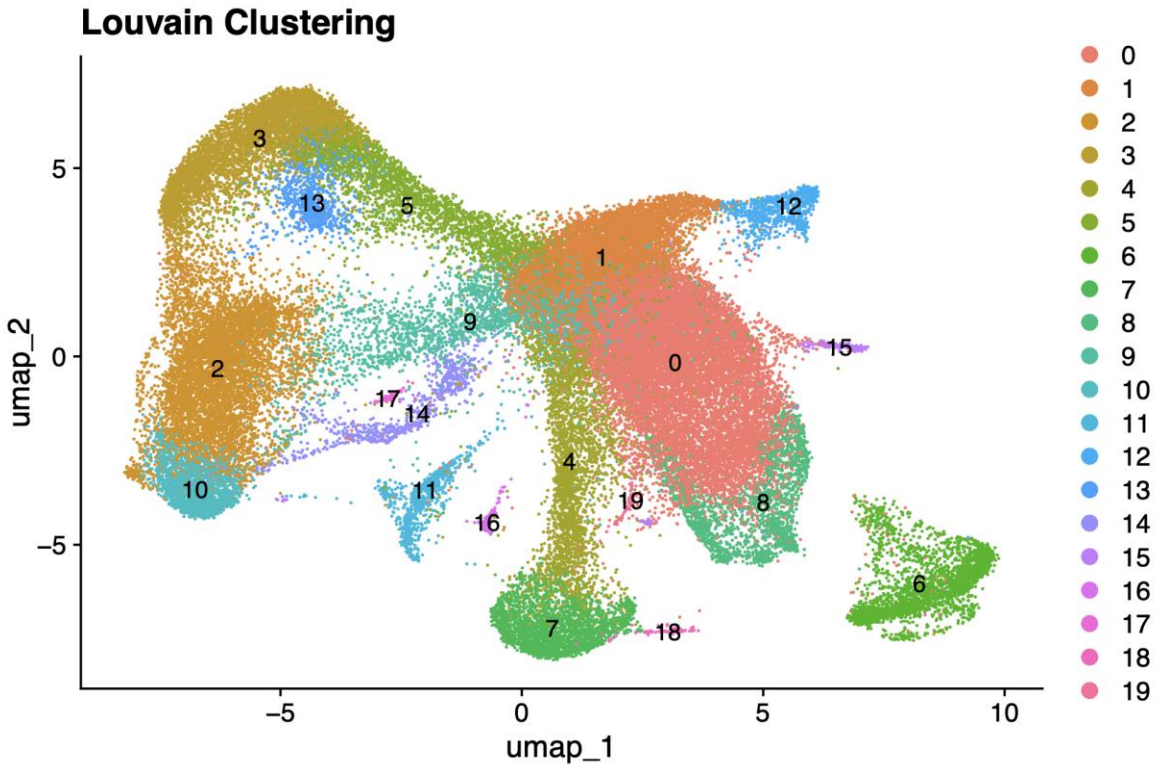


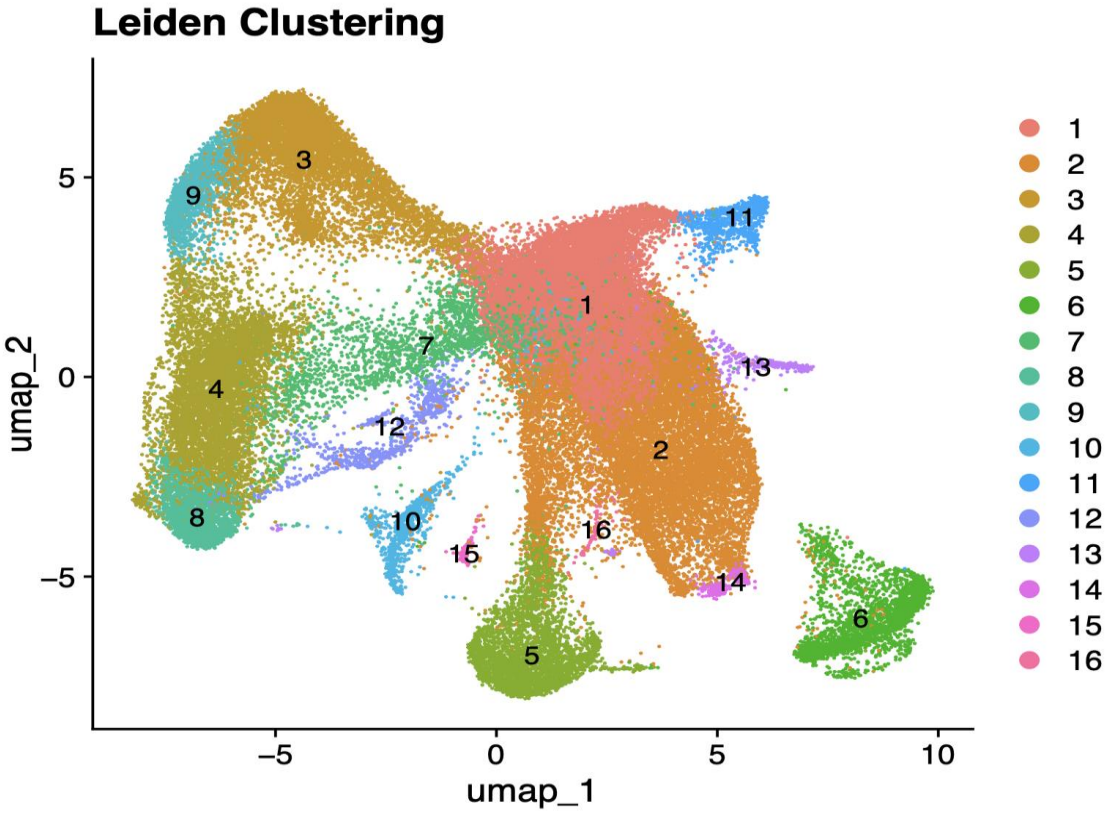
How did you choose the number of dimensions? Use a plot to explain. The number of dimensions selected were based on the PCA Elbow Plot. The Elbow Plot shows a clear inflection point around PC 15–20, where the variance explained by additional PCs rapidly decreases. PCs beyond this point mainly represent technical noise rather than biological structure. Therefore, we used the first 20 principal components for downstream UMAP and clustering.

Explain why we use a combination of PCA with UMAP for clustering and not only one of the methods.

PCA followed by UMAP was used because PCA reduces the high-dimensional gene expression matrix to a denoised, low-dimensional representation. PCA removes technical noise and preserves global variance structure. UMAP is then applied to these principal components to capture non-linear relationships and preserve local neighborhoods in the data. UMAP alone on raw counts is unstable and noisy, while PCA alone cannot capture complex biological structure. Therefore, PCA → UMAP combines the strengths of both methods and produces a more accurate and interpretable embedding.

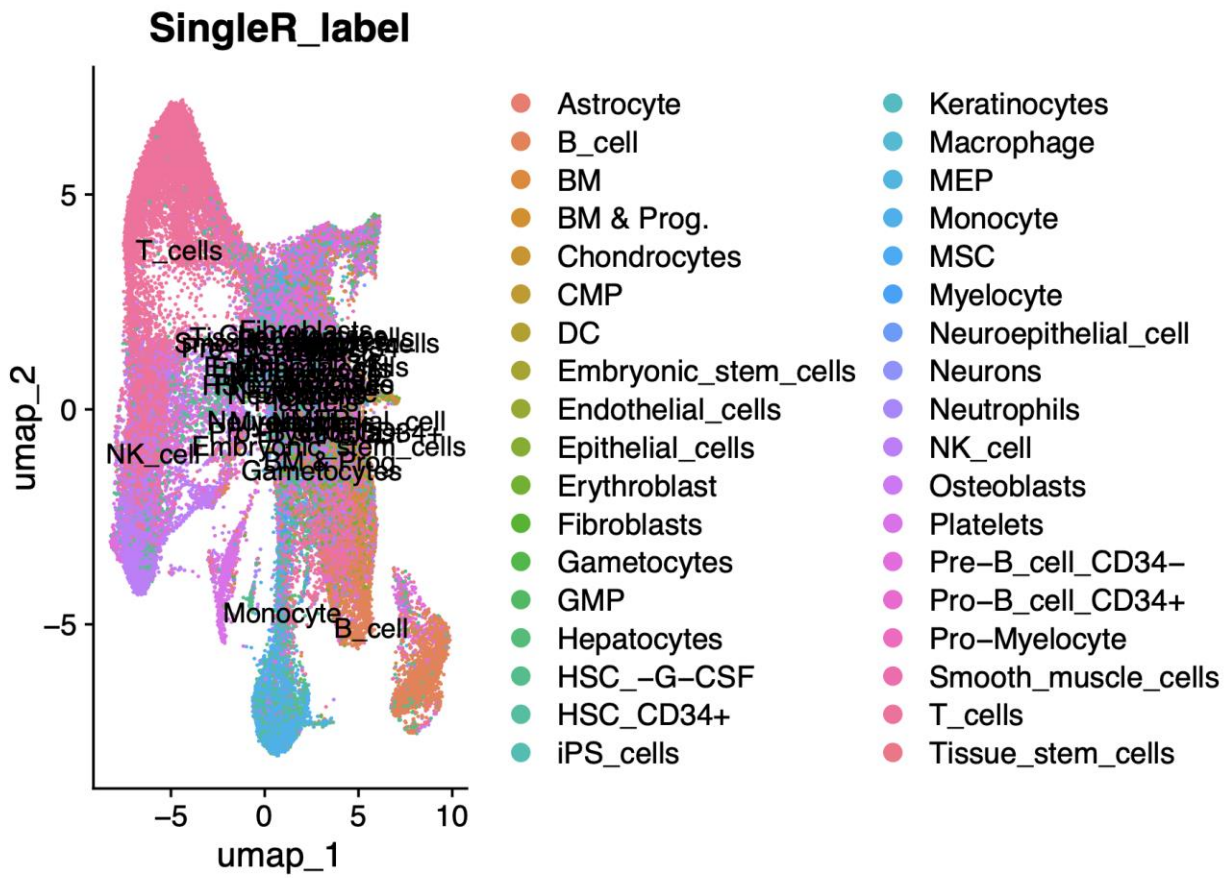
5.2 Clustering

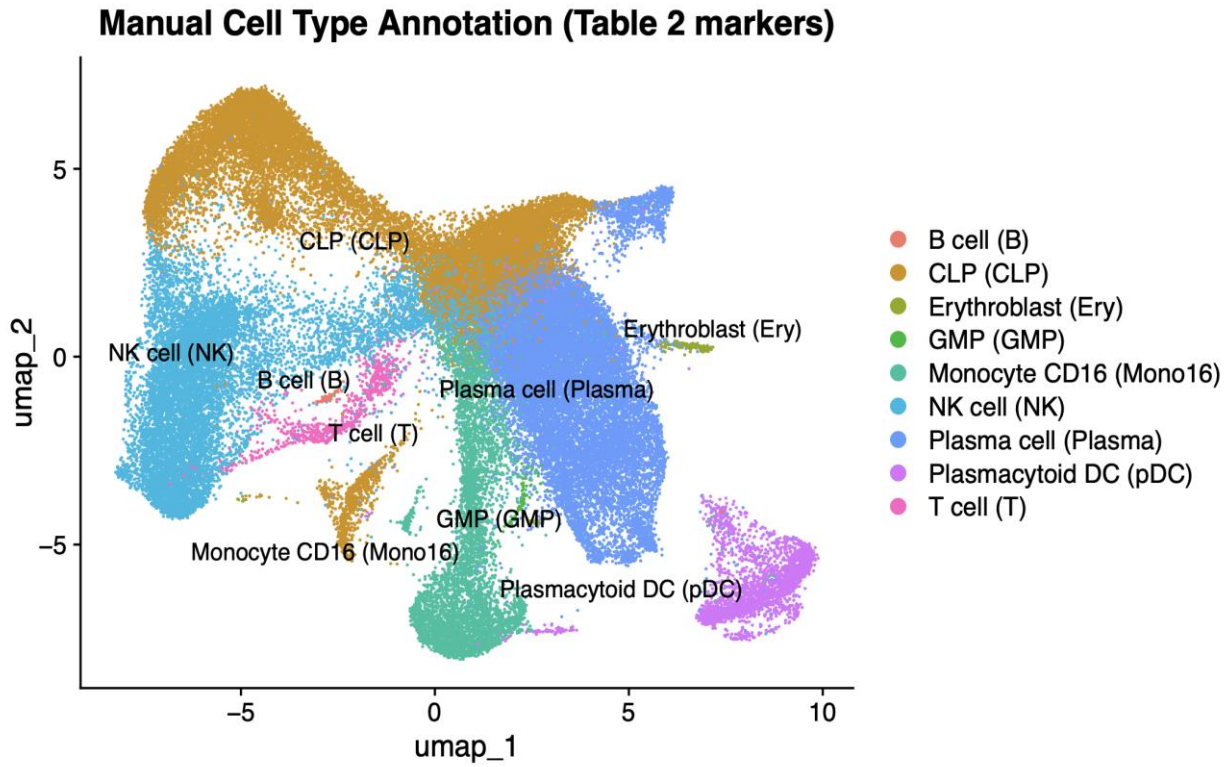


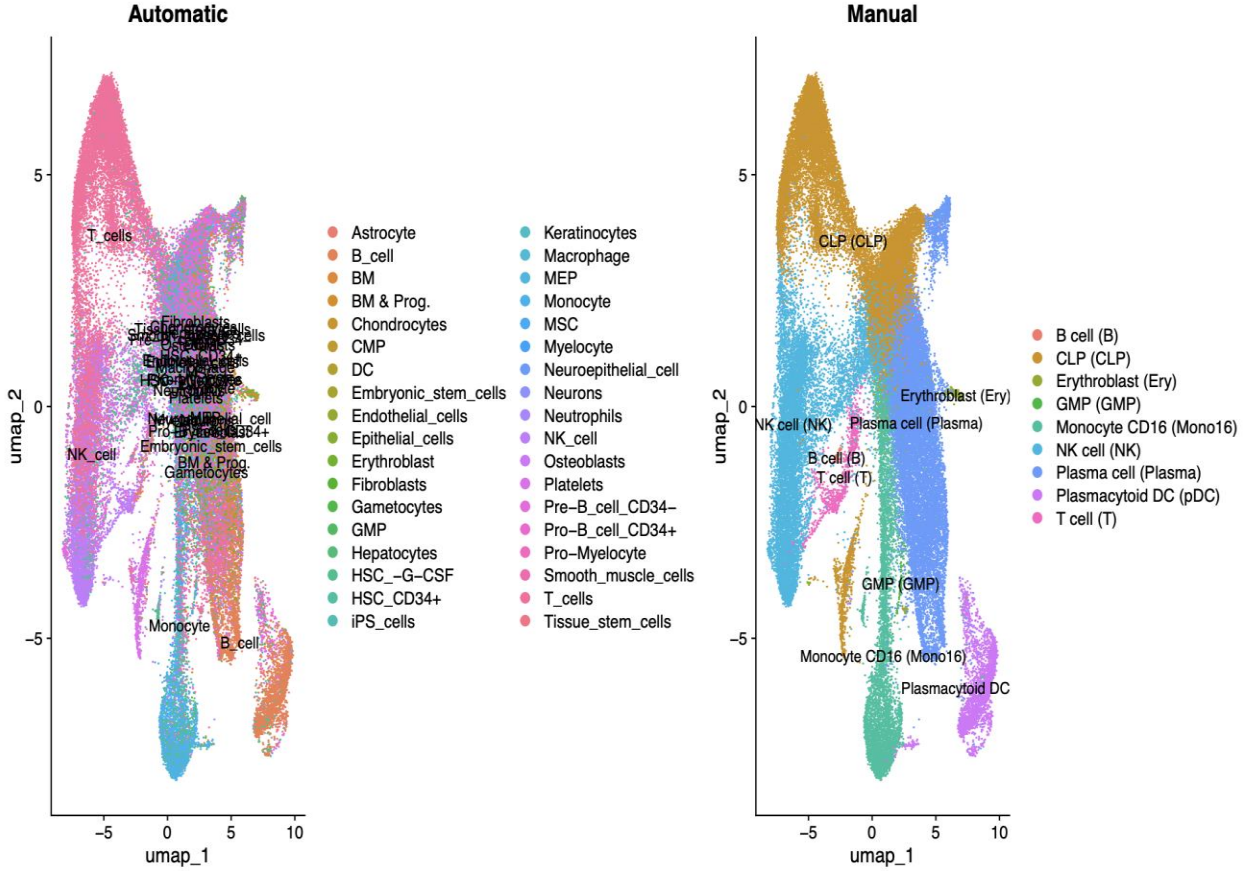


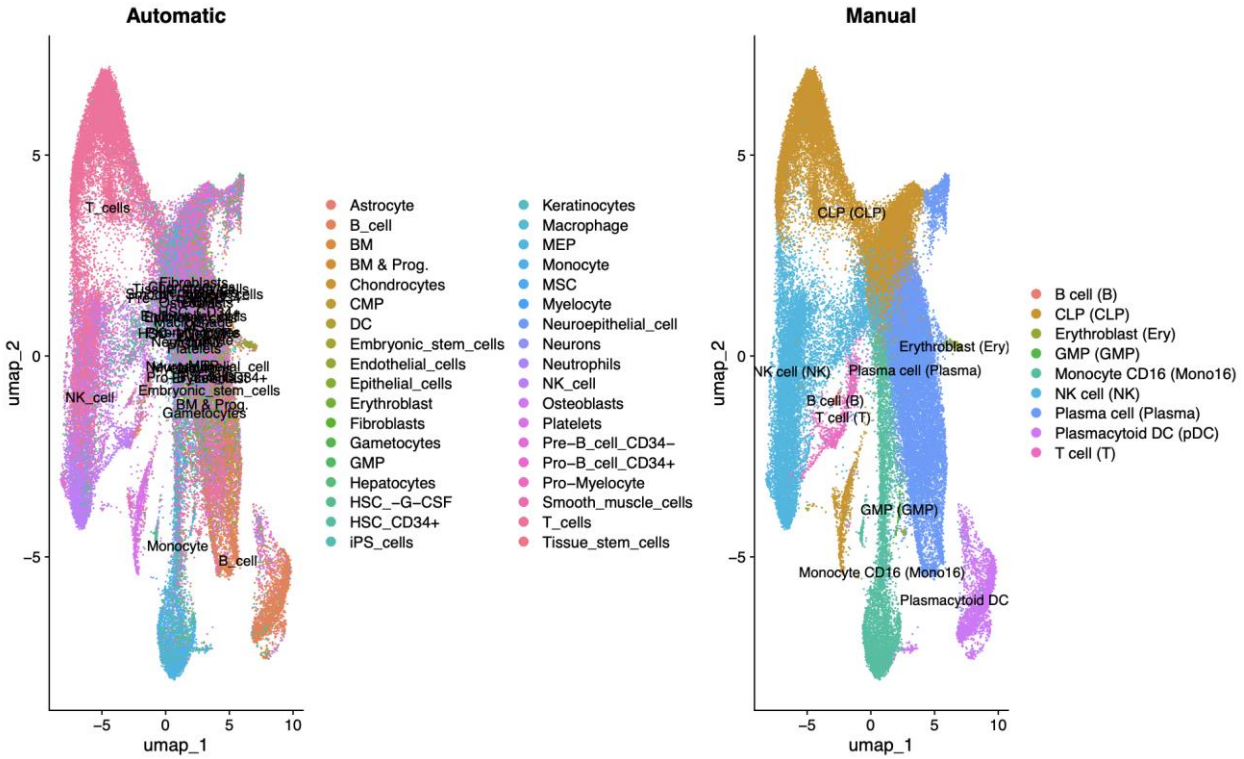
Do you observe any differences in Leiden and Louvian clustering?
Yes, there are clear differences between Leiden and Louvain clustering.
Leiden identifies more refined and well-separated clusters, especially splitting large Louvain clusters into smaller subclusters. It also better resolves rare and transitional cell populations, producing sharper cluster boundaries on the UMAP. Louvain tends to form broader, more merged clusters, whereas Leiden gives a more granular and biologically meaningful structure.

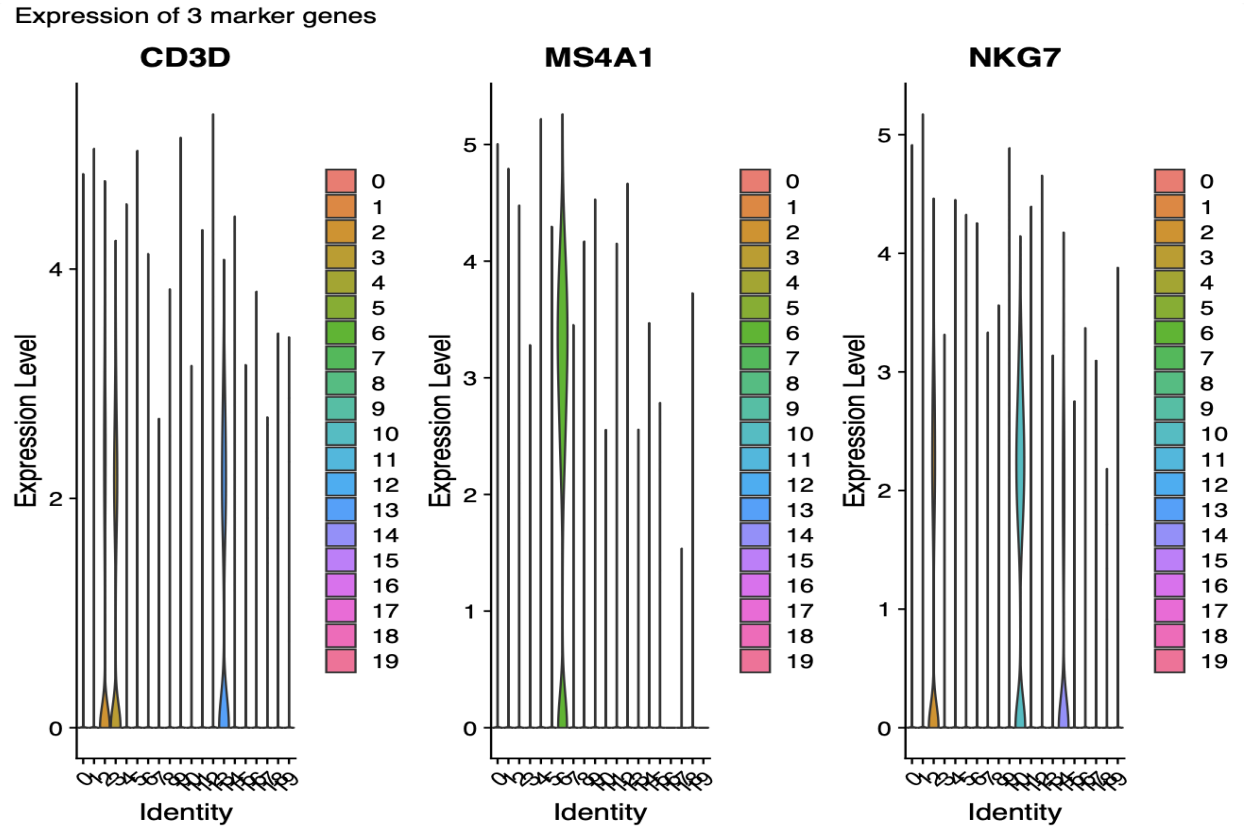
Week 3

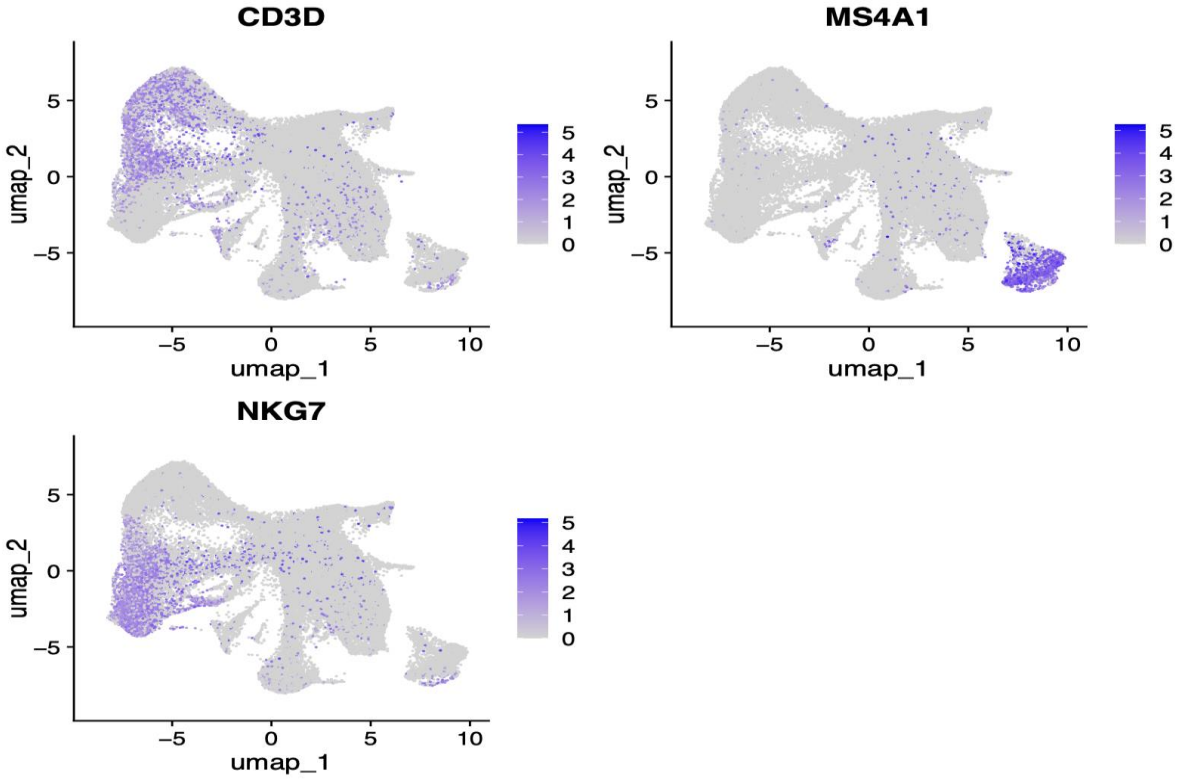



Manual Cell Type Annotation (Table 2 markers)
CLP (CLP)
Erythroblast (Ery)
NK cell (NK)
B cell (B)
Plasma cell (Plasma)
T cell (T)
GMP (GMP)
Monocyte CD16 (Mono16)
Plasmacytoid DC (pDC)
umap_1
umap_2
B cell (B)
CLP (CLP)
Erythroblast (Ery)
GMP (GMP)
Monocyte CD16 (Mono16)
NK cell (NK)
Plasma cell (Plasma)
Plasmacytoid DC (pDC)
T cell (T)

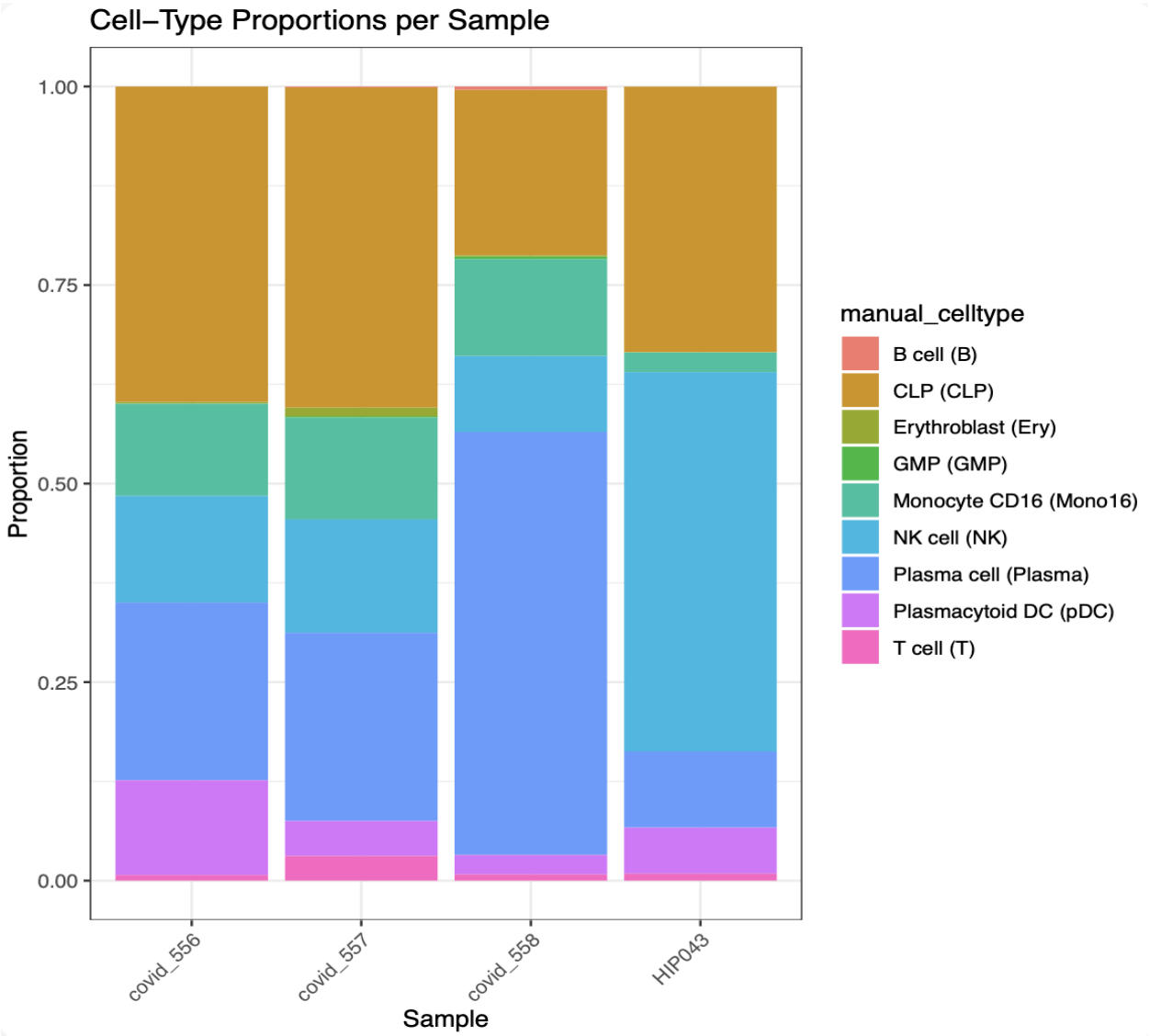

Automatic
Manual
umap_1
umap_2
Astrocyte
B_cell
BM
BM & Prog.
Chondrocytes
CMP
DC
Embryonic_stem_cells
Endothelial_cells
Epithelial_cells
Erythroblast
Fibroblasts
Gametocytes
GMP
Hepatocytes
HSC_-G-CSF
HSC_CD34+
iPS_cells
Keratinocytes
Macrophage
MEP
Monocyte
MSC
Myelocyte
Neuroepithelial_cell
Neurons
Neutrophils
NK_cell
Osteoblasts
Platelets
Pre-B_cell_CD34-
Pro-B_cell_CD34+
Pro-Myelocyte
Smooth_muscle_cells
T_cells
Tissue_stem_cells
B cell (B)
CLP (CLP)
Erythroblast (Ery)
GMP (GMP)
Monocyte CD16 (Mono16)
NK cell (NK)
Plasma cell (Plasma)
Plasmacytoid DC (pDC)
T cell (T)


Automatic
Manual
umap_1
umap_2
T_cells
NK_cell
Monocyte
B_cell
CLP (CLP)
Erythroblast (Ery)
Plasma cell (Plasma)
B cell (B)
T cell (T)
GMP (GMP)
Monocyte CD16 (Mono16)
Plasmacytoid DC
Astrocyte
B_cell
BM
BM & Prog.
Chondrocytes
CMP
DC
Embryonic_stem_cells
Endothelial_cells
Epithelial_cells
Erythroblast
Fibroblasts
Gametocytes
GMP
Hepatocytes
HSC_-G-CSF
HSC_CD34+
iPS_cells
Keratinocytes
Macrophage
MEP
Monocyte
MSC
Myelocyte
Neuroepithelial_cell
Neurons
Neutrophils
NK_cell
Osteoblasts
Platelets
Pre-B_cell_CD34-
Pro-B_cell_CD34+
Pro-Myelocyte
Smooth_muscle_cells
T_cells
Tissue_stem_cells
B cell (B)
CLP (CLP)
Erythroblast (Ery)
GMP (GMP)
Monocyte CD16 (Mono16)
NK cell (NK)
Plasma cell (Plasma)
Plasmacytoid DC (pDC)
T cell (T)


Expression of 3 marker genes
CD3D
MS4A1
NKG7
Expression Level
Identity
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19


CD3D
MS4A1
NKG7
umap_1
umap_2
5
0
−5
10
5
4
3
2
1
0

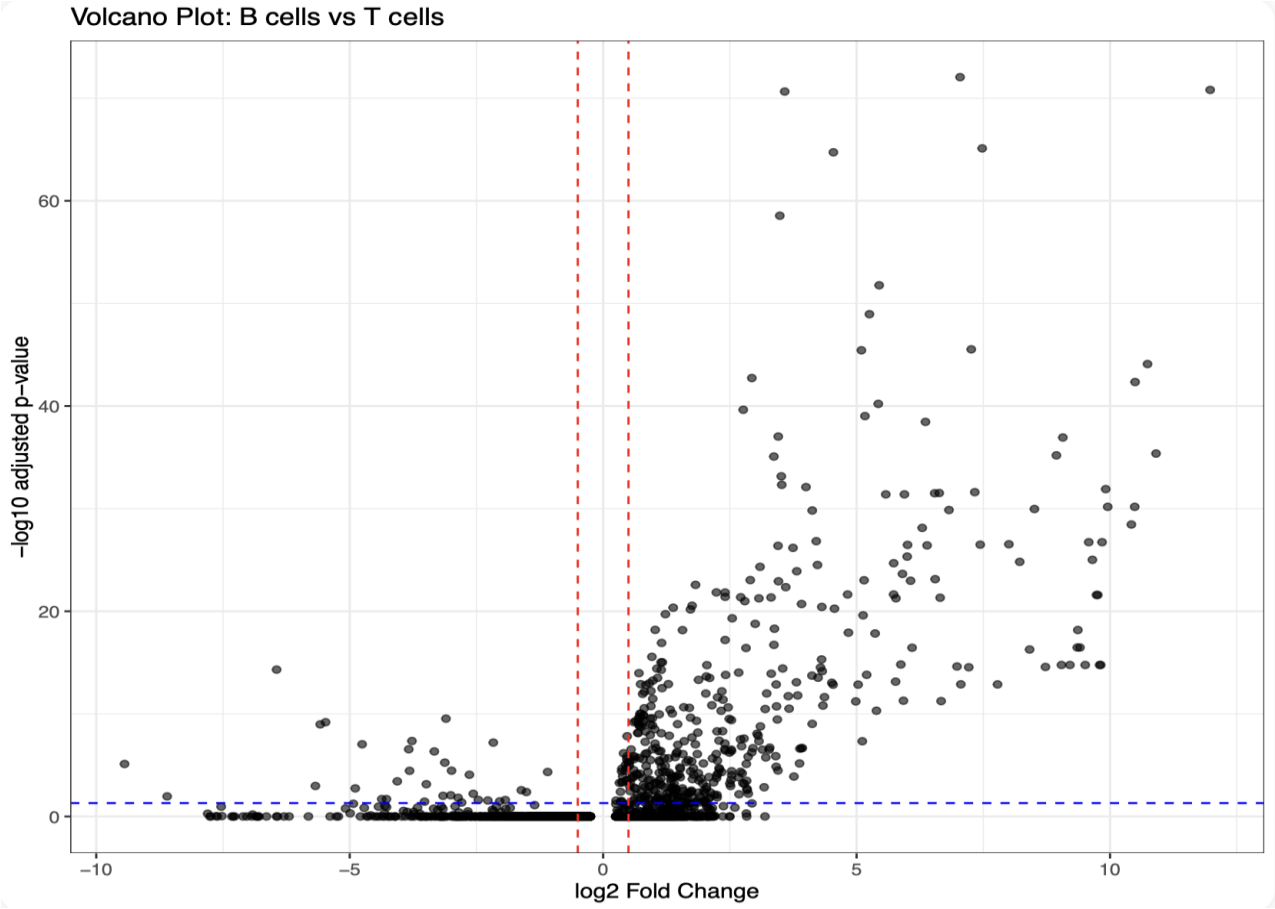
6.3 Cell-Type Proportions

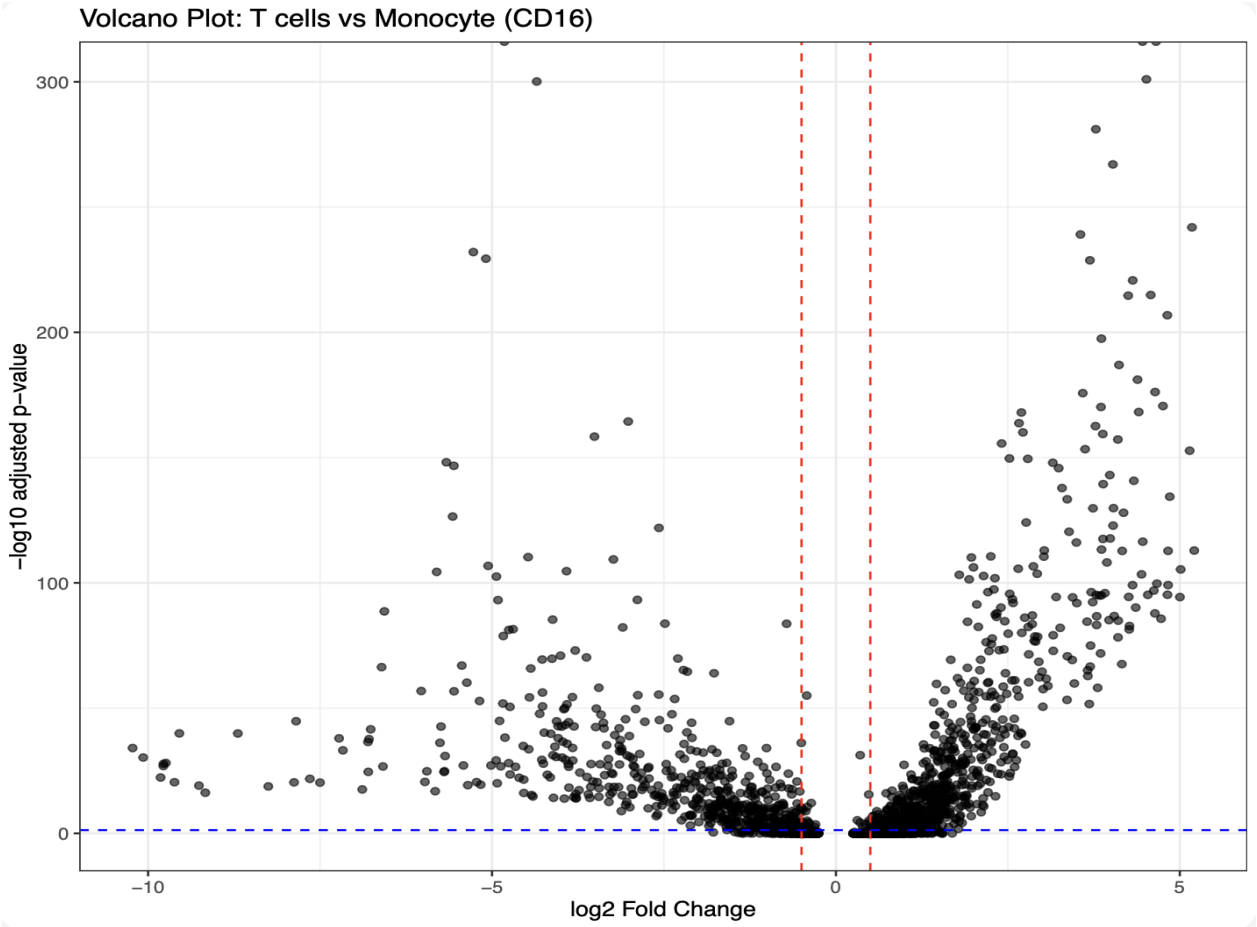


Explain how samples vary in terms of cell-type proportions.

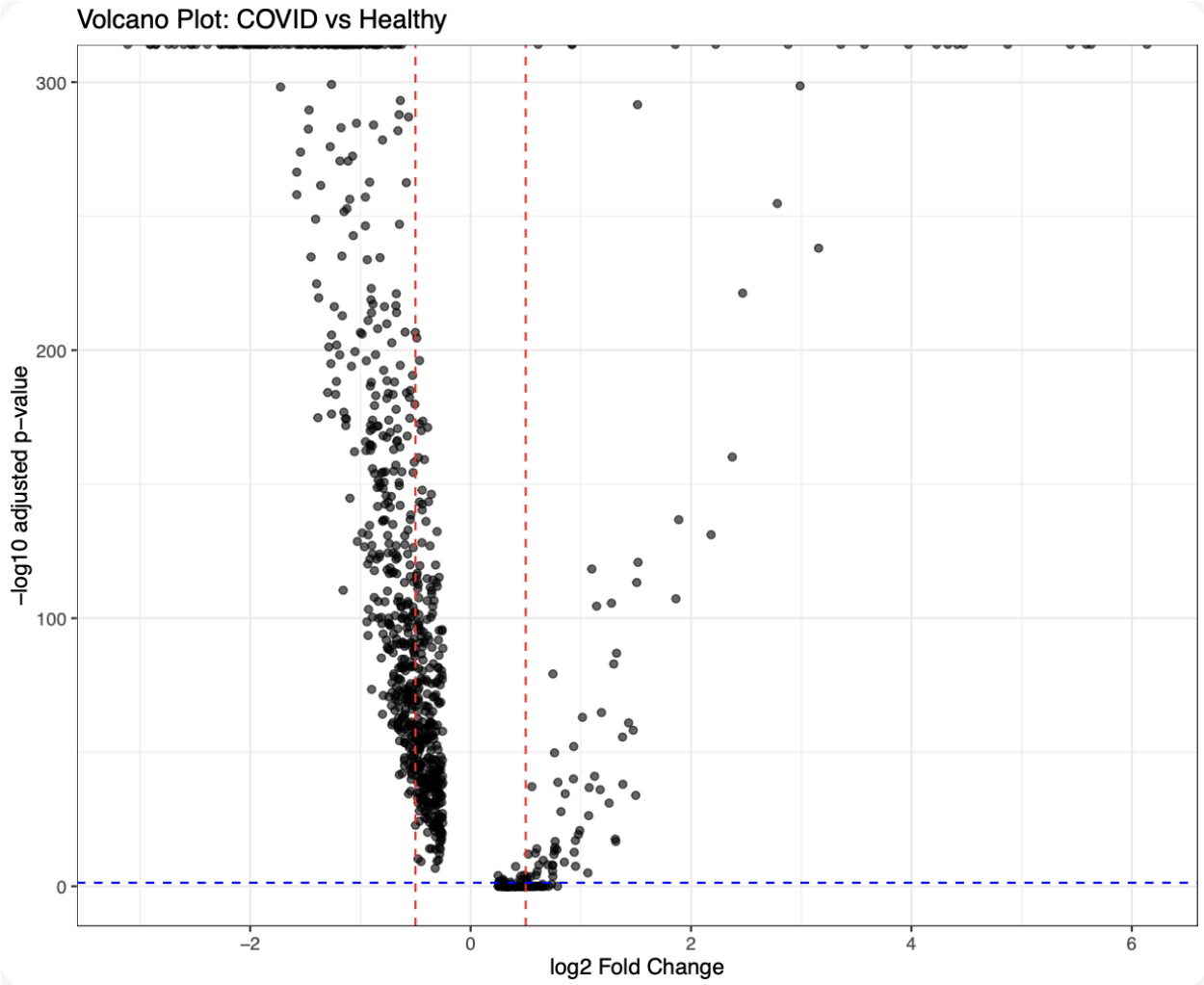
Cell-type proportions show clear sample-specific differences. HIP043 is dominated by NK cells (~48%), indicating a cytotoxic lymphocyte rich profile, while the COVID samples have substantially fewer NK cells. Both covid_556 and covid_557 show similar profiles, with CLP cells being the largest population (~40%), followed by moderate levels of NK cells, Mono16 monocytes, and plasma cells. In contrast, covid_558 is distinctly different, being dominated by plasma cells (~56%) and having very few NK cells. This suggests strong plasmablast activation in this sample. Small populations of T cells, B cells, erythroblasts, and pDCs are present in all samples but vary in relative proportions.

7.1 Differential Expression Analysis on cell-types




Volcano Plot: T cells vs Monocyte (CD16)
−log10 adjusted p−value
300
200
100
0
−10
−5
0
5
log2 Fold Change

7.1.1 Memory Formation on Cells



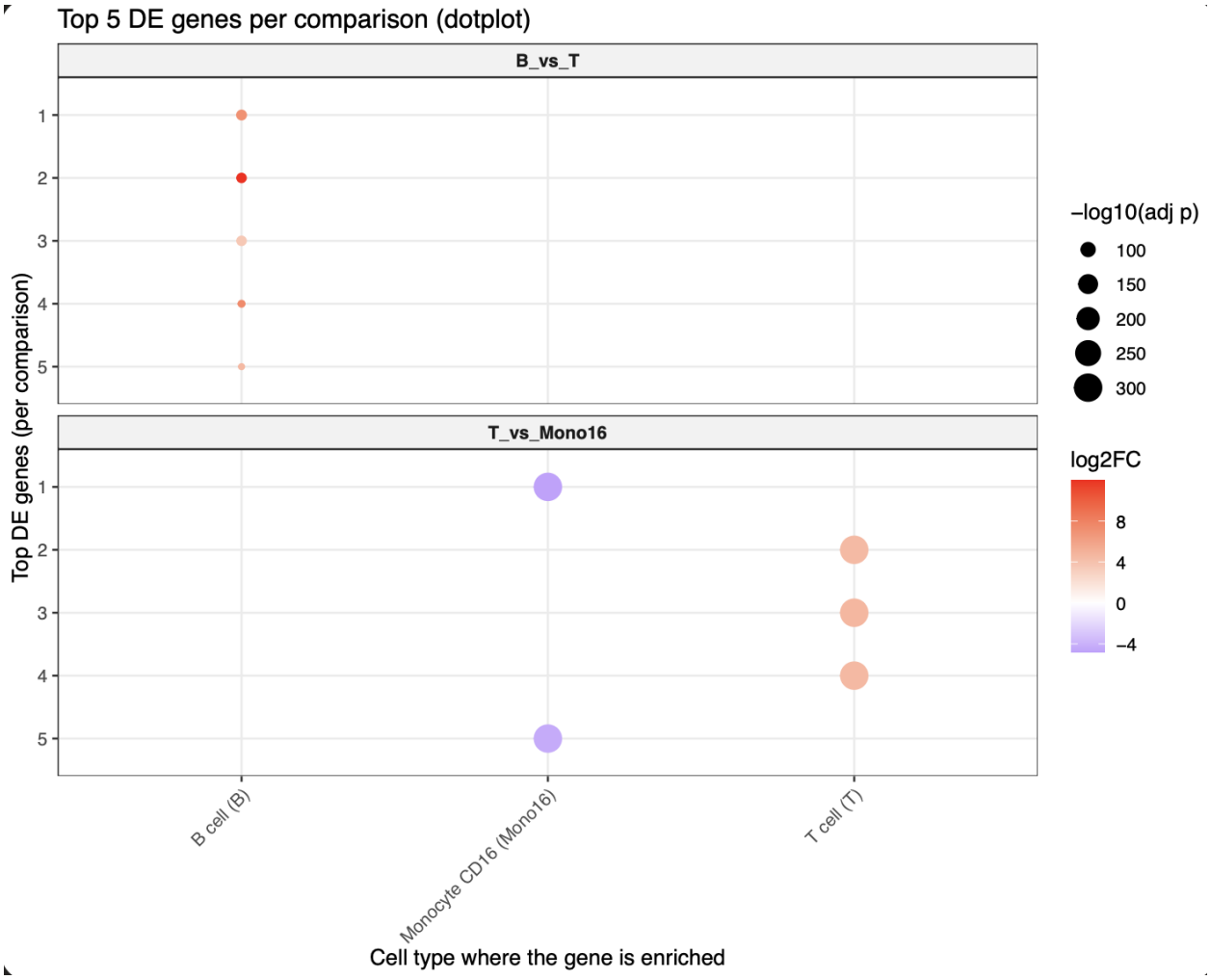
What are the differential genes you observed?

The main differential genes between COVID-19 and healthy samples include strong upregulation of interferon-responsive genes (e.g., IFI27), inflammatory myeloid markers (S100A8, S100A9, LYZ), and multiple immunoglobulin genes (IGHG1, IGHG4, IGHM). Conversely, cytotoxic and NK/T-cell–associated genes such as GNLY, KLRB1, SH2D1B, and TRDC are significantly downregulated. These patterns reflect increased inflammatory and humoral responses alongside reduced cytotoxic immune activity in COVID-19 cells.

```
> top_up
       p_val avg_log2FC pct.1 pct.2 p_val_adj   gene
IGHG4      0   6.137230 0.272 0.007         0  IGHG4
IGHG1      0   5.630232 0.389 0.019         0  IGHG1
IFI27      0   5.586859 0.152 0.002         0  IFI27
S100A8     0   5.442406 0.377 0.013         0 S100A8
S100A9     0   4.873907 0.367 0.016         0 S100A9
LYZ        0   4.472554 0.243 0.011         0    LYZ
VCAN       0   4.416010 0.242 0.013         0   VCAN
IGHG3      0   4.331459 0.208 0.013         0  IGHG3
IGHG2      0   4.227610 0.260 0.019         0  IGHG2
IGHM       0   3.972983 0.507 0.067         0   IGHM

> top_down
        p_val avg_log2FC pct.1 pct.2 p_val_adj    gene
SH2D1B      0  -3.111311 0.009 0.102         0  SH2D1B
PRR20C      0  -2.909004 0.011 0.105         0  PRR20C
SPON2       0  -2.903886 0.036 0.260         0   SPON2
KLRB1       0  -2.855625 0.021 0.145         0   KLRB1
GNLY        0  -2.854758 0.106 0.506         0    GNLY
SLC38A7     0  -2.741411 0.056 0.334         0 SLC38A7
TRDC        0  -2.685720 0.013 0.103         0    TRDC
S1PR5       0  -2.601873 0.030 0.205         0   S1PR5
MATK        0  -2.536227 0.023 0.155         0    MATK
FCGR3A      0  -2.531678 0.075 0.385         0  FCGR3A
```

7.2 Plot Differentially expressed genes

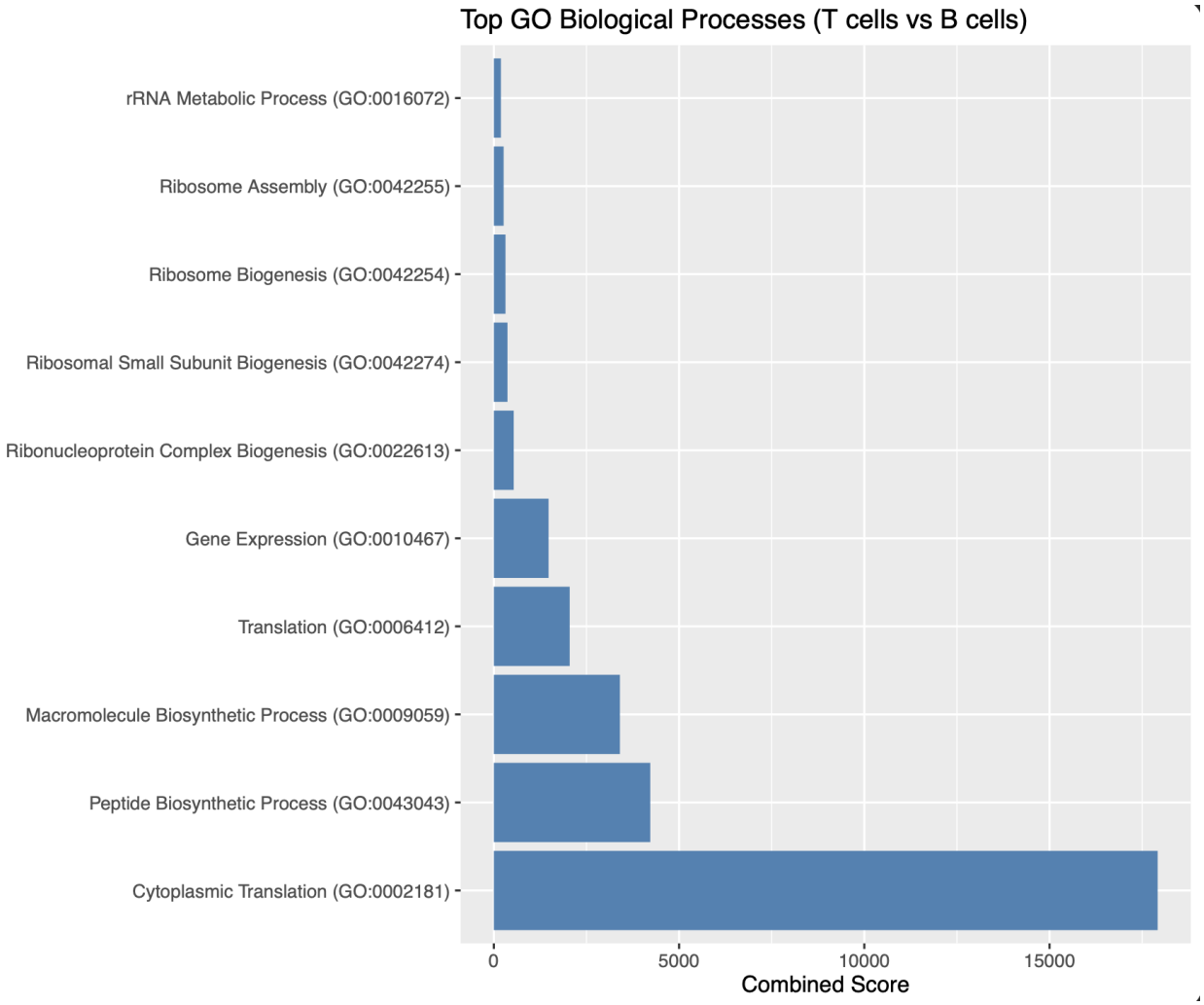


8.1 Differential Expression Analysis on groups

Report the top 5 DEGs with p-value and Fold change.
Differential expression analysis comparing T cells versus B cells identified the following top 5 most significant DEGs (ranked by adjusted p-value):

Gene	log2 Fold Change	Adjusted p-value
RP5-1028K7.2	-7.04	8.83×10^{-73}
RASD2	-11.98	1.55×10^{-71}
CDK6	-3.58	2.25×10^{-71}
LAPTM4B	-7.48	7.75×10^{-66}
SOX4	-4.54	1.87×10^{-65}

8.2 Pathway analysis on groups



8.3 Biological interpretation

Name the pathway with the lowest p-value. Explain its biological meaning.
The pathway with the lowest adjusted p-value was Cytoplasmic Translation (GO:0002181). This pathway represents the synthesis of proteins by cytoplasmic ribosomes and includes ribosomal structural proteins, translation factors, and rRNA processing components.
This result indicates that B cells in our dataset have significantly higher activity in ribosomal and translational pathways compared to T cells. Biologically, this is expected because B cells—especially antibody-producing plasma cells—require very high levels of protein synthesis to produce immunoglobulins. Therefore, the enrichment of translation-related pathways reflects the higher biosynthetic and metabolic activity characteristic of B-cell function.

Week 4

9.1 Identifying Informative Clusters

Which subset of clusters would you select for trajectory analysis, and why?
I would select clusters C1, C2, C3, C4, C5, and C6 for trajectory analysis. These 6 clusters together form the hypothesised developmental axis. C1–C2 represent stem/progenitor states (candidate root), C3–C4 are intermediate/transitional states, and C5–C6 are terminal differentiated end points (two terminal lineages). Including all of them allows the trajectory method to recover a single linear progression or a branching structure with progenitor → intermediate → terminal transitions. Excluding any of these stages would remove information about either the root, the intermediate transition, or terminal identity and could prevent correct ordering or detection of branching.

Under what conditions might you exclude a cluster?
I might exclude a cluster only when there is clear evidence that it would confound trajectory inference. Typical, justifiable reasons are:
1) Technical artefact or doublets: the cluster is enriched for doublet signatures, abnormally high UMI counts, or QC metrics consistent with artefact.
2) Strong batch or donor effect: cluster membership is driven by batch/donor rather than biological variation and cannot be corrected.
3) Very low cell number: the cluster contains too few cells to estimate expression trends reliably (practically: << 20–50 cells depending on method).
4) Unrelated lineage / contamination: marker analysis shows lineage-defining markers that are incompatible with the investigators' developmental hypothesis (e.g., epithelial markers while others are neural).

5) Discrete, non-transitional profile: the cluster does not show any intermediate expression between neighbouring clusters (no mixed marker expression, no gradual change), suggesting it is not part of the same continuum.
When excluding, one should record and report the reason and consider a sensitivity analysis (trajectory with and without the cluster).

How would marker gene expression patterns help confirm these clusters belong to a single developmental continuum rather than unrelated processes?
Using the following diagnostic criteria and visual checks; concordance across multiple criteria supports a single developmental continuum:
1) Ordered gradients: the expression levels of stem/progenitor markers (such as SOX2 and NANOG) should be highest in clusters C1 and C2, gradually decreasing across C3, C4, and reaching the lowest levels in C5 and C6, while terminal lineage markers should progressively increase toward C5 or C6.
2) Coexpression in intermediates: in intermediate clusters (C3 and C4), there should be coexpression of both early (stem) and late (lineage) markers at moderate levels, indicating transitional programs rather than distinct, opposing expressions.
3) Monotonic gene pseudotime correlations: a number of genes should demonstrate significant correlations (whether positive or negative) with the inferred pseudotime, leading to smooth expression trends when graphed against pseudotime, without random fluctuations.
4) Directional RNA velocity: the RNA velocity vectors should orient from clusters C1 and C2 toward C3 and C4, progressing to C5 and C6, thereby confirming the suggested direction of differentiation.
5) Shared pathway progression: the activity of gene sets or pathways should change in a systematic manner, such as a decline in proliferative or biosynthesis programs while lineage-specific functional programs are upregulated.
6) Continuous embedding: in reduced dimensions (diffusion map / PCA / UMAP restricted to selected clusters), cells should form a continuous manifold or ordered arc connecting progenitors to terminals rather than separate disconnected islands.
If the above checks hold (ordered marker gradients, transitional coexpression, monotonic pseudotime trends, consistent velocity direction, and continuous embedding), the clusters are very likely part of a single developmental continuum. If these checks fail (discrete, conflicting marker signatures, no gradient or inconsistent velocity), the clusters are more likely unrelated processes and should be reconsidered for exclusion or reannotation.

9.2 Root Node Selection

Which of C1 or C2 would you select as a root node, and what biological principles support that choice?

I would select as root the cluster (C1 or C2) that best meets progenitor criteria:

(a) Highest levels of stem/progenitor markers (such as SOX2 and NANOG) compared to the other clusters

(b) Low or negligible levels of differentiation markers that characterize C5/C6

(c) Increased signals of proliferation (cell-cycle-related genes, ribosomal/proliferation signatures) and/or greater transcriptional entropy, both characteristic of stem/progenitor states

(d) RNA velocity vectors directed away from that cluster towards C3/C4 and subsequently to C5/C6

Biologically, the root should be the population that most closely resembles the known earliest stage in the lineage hierarchy (marker-wise and functionally) and from which directional transcriptomic change is observed. If C1 shows stronger stem marker expression and outward velocities while C2 already co-expresses intermediate markers, choose C1 as root (and vice versa if the evidence points to C2).

Could both C1 and C2 serve as valid root candidates? How would choosing C1 versus C2 shift pseudo-time ordering and branching structure?

Yes both can be valid if they are both bona fide progenitor subpopulations (e.g., two stem-like states or minor/major progenitor subsets). Differences when choosing one vs the other:

Pseudo-time origin and scaling: pseudo-time will be anchored to the chosen root, so cells in the other progenitor cluster will appear earlier or slightly downstream depending on relative transcriptional distance. Choosing C1 makes trajectories start at C1 and flow through C2→C3/4→C5/6 (if C2 is slightly downstream), while choosing C2 reverses that local ordering.

Branching topology sensitivity: the inferred branching point (where the path splits toward C5 vs C6) may shift upstream or downstream by the distance between C1 and C2; branch lengths and the apparent timing of fate decisions can change.

Interpretation of intermediates: intermediate clusters (C3/C4) may be placed at different pseudo-time positions relative to the root, which can alter which genes are identified as early vs mid regulators.

In practice, if both are plausible roots, one should (a) test both choices, (b) check consistency of key marker trends and velocity directions, and (c) report how branch location and top DE genes change with root selection.

Suppose a root node is incorrectly chosen in C5. How would this influence: (a) the direction of pseudo-time, (b) the interpretation of C1 as early or late, and (c) the inferred developmental path?

(a) Direction of pseudo-time: it will be reversed or scrambled, pseudo-time will start at a terminal state and “flow” toward progenitors, producing an inverted ordering of biological progression.

(b) Interpretation of C1 (early vs late): C1 (true progenitor) would be interpreted as late in pseudo-time (i.e., downstream), leading to the false conclusion that progenitor programs arise after terminal programs; marker trends will appear inverted (stem markers increasing with pseudo-time rather than decreasing).

(c) Inferred developmental path: the trajectory will be biologically misleading: branch points and timing of fate decisions will be inverted, DE genes identified along pseudo-time will reflect reverse transitions (which are not real differentiation events), and downstream analyses (e.g., gene regulation, pathway timing, fate bias) will be confounded. RNA velocity and known marker gradients will conflict with the pseudo-time, flagging the root choice as incorrect.

Conclusion: choosing a terminal cluster as root produces systematically inverted and biologically implausible results, always validate root selection with marker gradients, velocity, and known biology before relying on downstream interpretations.

9.3 Algorithmic vs Biological Reasoning

What assumptions do Monocle, Slingshot, or similar tools make?

These trajectory inference tools typically assume that:

1) Cells follow a smooth and continuous transcriptional progression from progenitor to terminal states.

2) High-dimensional gene expression changes can be represented as a low-dimensional manifold with coherent transitions.

3) Intermediate states lie between early and late populations in expression space, forming a connected graph structure.

4) Branch points correspond to biological lineage bifurcations, not technical or batch-driven variation.

5) Sampling is dense enough to capture all major intermediate states, without large gaps.

Can these assumptions fail in real datasets? Give examples.

Yes, these assumptions can fail under several real-world conditions:

1) Batch effects or donor differences create artificial separations, making unrelated clusters appear disconnected.

2) Transient or rare cell states may be under-sampled, breaking the continuity required for trajectory inference.

3) Convergent differentiation (different lineages producing similar mature states) can falsely appear as a single lineage.

4) Cell-cycle effects or stress responses may dominate variation, misleading algorithms into inferring incorrect pseudo-time paths.

5) Cell populations may not represent a true developmental hierarchy, e.g., mixed immune activation states rather than lineage progression.

How would you validate a computationally inferred trajectory experimentally?

A trajectory can be validated experimentally by:

1) Lineage tracing, using barcoding or genetic labeling to track descendant cells over time.

2) Time-course RNA-seq or scRNA-seq, showing that predicted early → late transitions occur chronologically.

3) Flow cytometry or immunostaining of stage-specific markers to confirm ordered acquisition or loss of markers.

4) Perturbation experiments, such as knocking out key regulators to see whether predicted downstream states fail to appear.

10 Cell-Cell Communication

Identify two specific cell types that show increased communication in the HC compared to the PFC (based on red edges in panels a or b). Briefly describe whether this difference is driven more by number of interactions or interaction strength.

1) GABAergic neurons: In panels (a) and (b) GABAergic nodes show many red edges and several thick red edges, indicating that in the hippocampus they both participate in more ligand–receptor pairs (higher number of interactions) and those interactions tend to have greater communication strength. The effect is visible in both plots but is most pronounced in panel (b) (interaction strength).

2) Microglia: Microglial edges in panel (b) are noticeably red and relatively thick (stronger), while panel (a) shows a more moderate increase in the number of connections. This pattern indicates the HC > PFC difference for microglia is driven more by increased interaction strength than by a large change in the number of distinct LR pairs.

Identify one cell type that appears to act as a major communication hub in both tissues (e.g., high connectivity in panels c and d). Based on its role in the brain, propose a hypothesis for why this cell type maintains a central signaling position.

Glutamatergic neurons appear as a major hub in both panels (c) (PFC) and (d) (HC):. They have many incoming/outgoing edges and relatively thick edges, indicating high connectivity in both tissues.

Biological hypothesis: The majority of synaptic signalling is produced by excitatory (glutamatergic) neurones, which are closely linked to a variety of supporting cell types, including astrocytes, oligodendrocyte lineage, endothelial cells, and microglia. Glutamatergic neurones naturally occupy a central signalling role, coordinating neurotransmission, metabolic coupling, and local immune/homeostatic responses because they regulate network activity and need metabolic, trophic, and synaptic support. This allows them to maintain high connectivity across brain regions.

Compare the overall communication density in panels c vs d. Does one tissue appear globally more interconnected? Discuss one biological reason why different brain regions might show different levels of intercellular signaling activity.

Observation: Panel (d) (hippocampus) appears globally more interconnected than panel (c) (PFC): edges in (d) are denser and, in several node pairs, thicker than in (c). This suggests higher overall communication activity in the HC.

Biological reason: The hippocampus is a region with high synaptic plasticity and ongoing structural remodeling (e.g., activity-dependent synaptic change, adult neurogenesis in some subregions). These processes require intense, coordinated signaling among neurons, glia and vasculature (growth factors, chemokines, metabolic signals), which would increase both the number and strength of ligand–receptor interactions relative to a less plastic region.